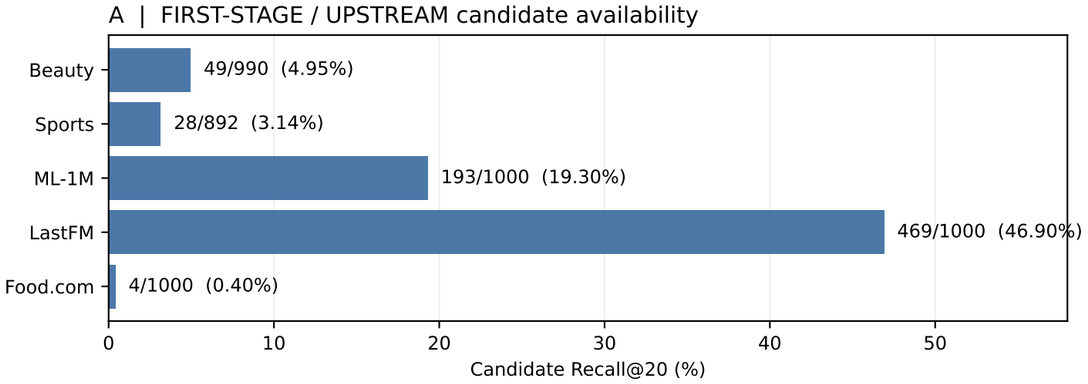
RQ7 | Two-stage candidate-aware re-ranking: availability and conditional utility

BPRMF / GRU4Rec / SASRec / LightGCN → shared top-20 pool → LLM re-ranking → final ranking

A | FIRST-STAGE / UPSTREAM candidate availability



D | Implemented single-positive denominator identity

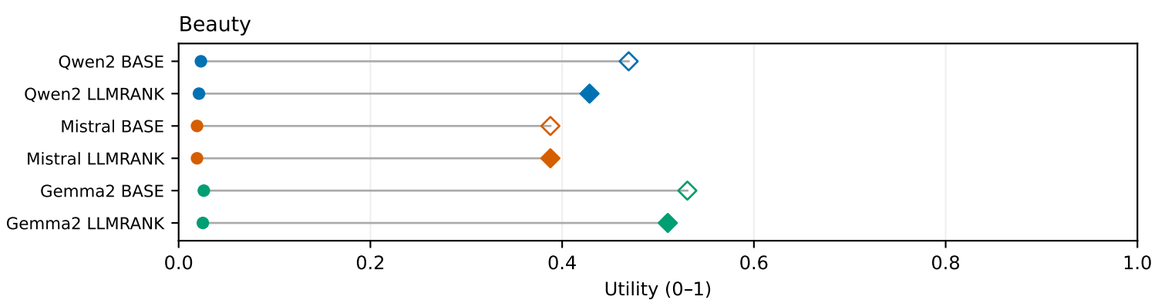
First-stage top-20 recall × LLM conditional utility
= all-user end-to-end utility

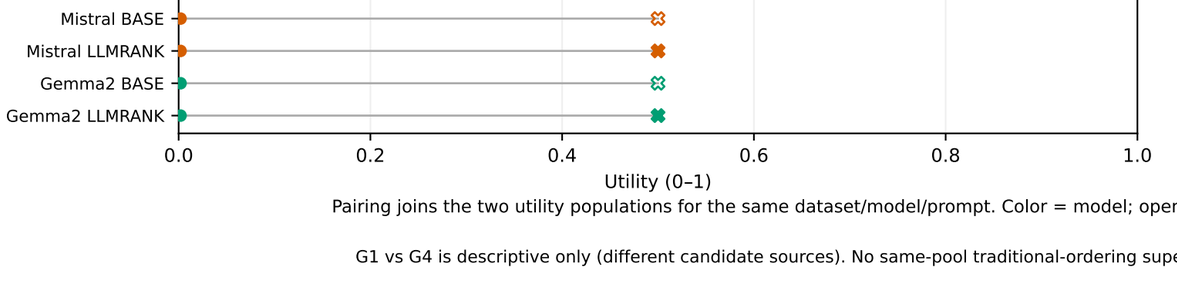
HR@5 e2e = Recall@20 × HR@5 | target present
NDCG@5 e2e = Recall@20 × NDCG@5 | target present

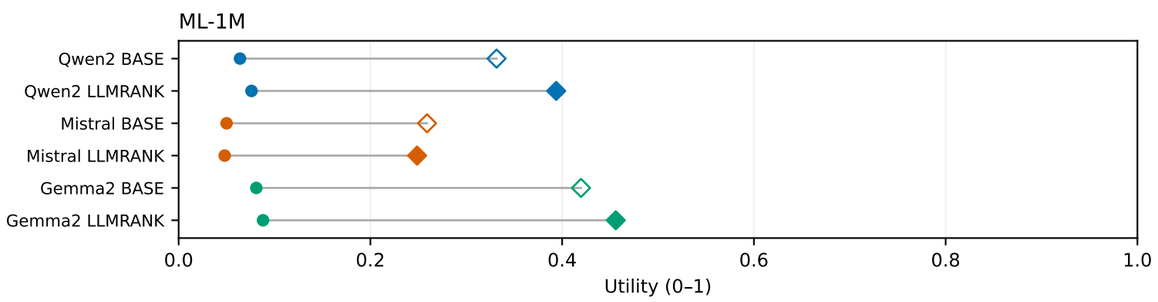
Verified in all 30 G4 cells. Max |error|: HR 5.55e-17; NDCG 2.78e-17.

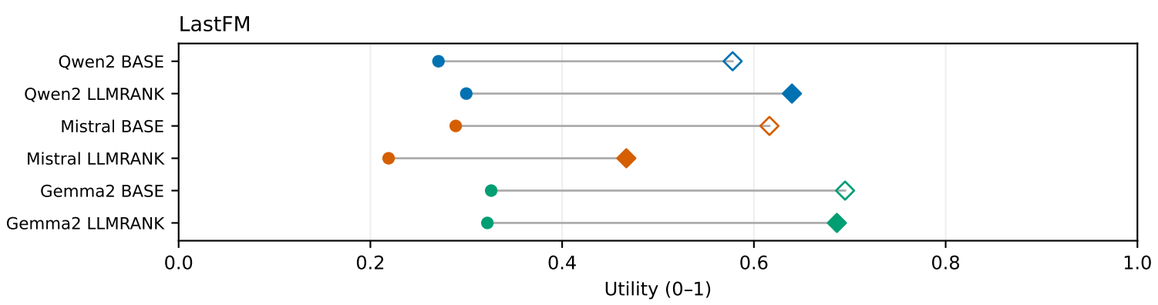
Recall belongs upstream, not to the LLM. Absent target and invalid output get zero credit.

B | HR@5: end-to-end ● to conditional ◆

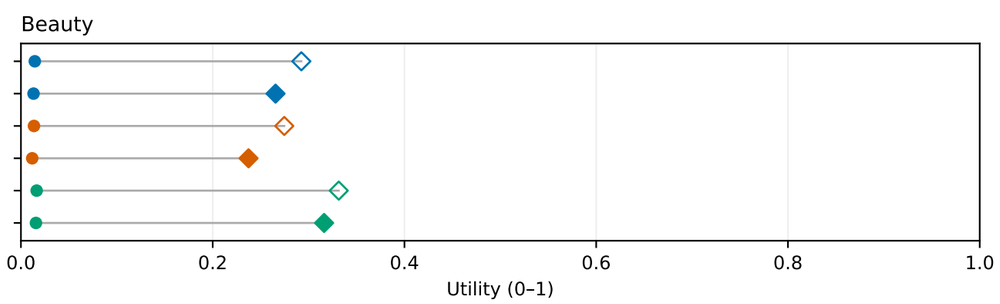


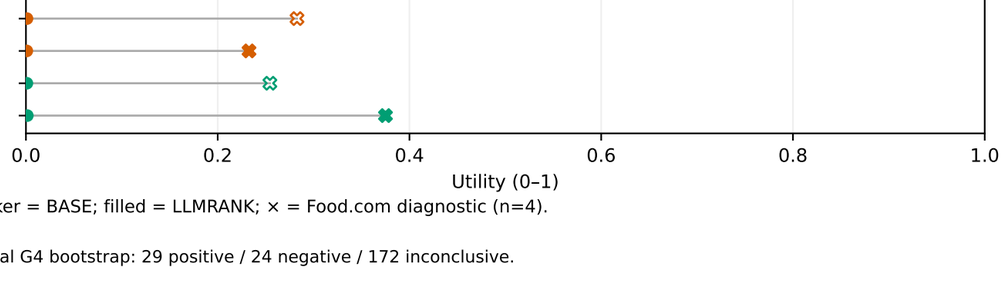


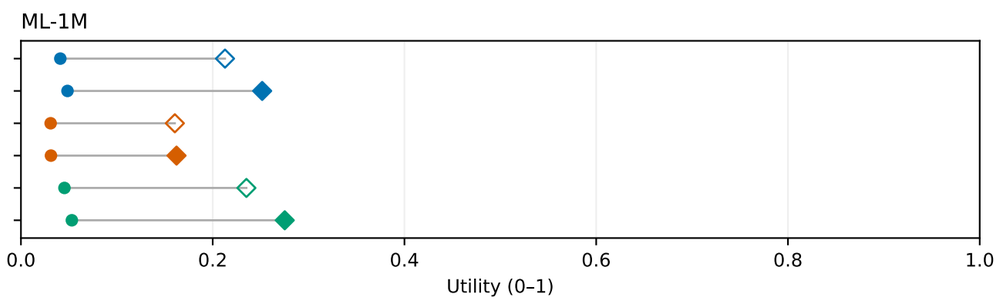


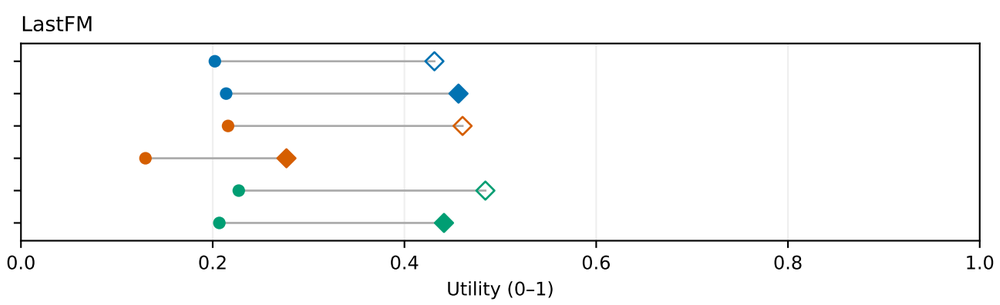


C | NDCG@5: end-to-end ● to conditional ◆









Pairing joins the two utility populations for the same dataset/model/prompt. Color = model; open conditional marker = BASE; filled = LLMRANK; × = Food.com diagnostic (n=4).

G1 vs G4 is descriptive only (different candidate sources). No same-pool traditional-ordering superiority test; internal G4 bootstrap: 29 positive / 24 negative / 172 inconclusive.